

Multiple Choice (10 marks)

1. Of the following page-replacement policies, which is guaranteed to incur the minimum number of page faults?
 - (a) Replace the page whose next reference will be the longest time in the future.
 - (b) Replace the page whose next reference will be the shortest time in the future.
 - (c) Replace the page whose most recent reference was the shortest time in the past.
 - (d) Replace the page whose most recent reference was the longest time in the past.
2. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
 - (a) 17_{16}
 - (b) $E4_{16}$
 - (c) $E7_{16}$
 - (d) $F4_{16}$
3. In Python 3, which of the following function removes all leading and trailing whitespace in string?
 - (a) `replace(old, new [, max])`
 - (b) `strip([chars])`
 - (c) `swapcase()`
 - (d) `title()`
4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address
5. Which of the following best describes the primary way in which a distributed denial-of-service (DDoS) attack differs from a denial-of-service (DoS) attack?
 - (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs

True / False (10 marks)

Answer True or False

6. True or False: Merge sort has a running time that is least affected by the initial ordering of the input compared to other sorting algorithms.
7. True or False: The access matrix approach, when stored directly, can be large and cumbersome to manage effectively.
8. True or False: Finding a shortest cycle in an undirected graph G can be solved in polynomial time.
9. True or False: Even after extensive testing, a Java program may still contain undetected bugs.
10. True or False: Python variable names are case-sensitive, meaning 'Variable' and 'variable' are treated as distinct identifiers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the second smallest number in a list.
12. Write a function to check whether the given month number contains 28 days or not.

End of Paper